DIGNITAS PS05 — 5-MINUTE TEAM PRESENTATION SCRIPT

Exact Timing Guide for 3 Speakers • Problem PS05: Adaptive Authentication & Real-Time Threat Mitigation

Speaker 1 0:00 – 1:35 (95 sec) Architecture & 5 Tiers	Speaker 2 1:35 – 3:15 (100 sec) 5 Threat Vectors & Bank vs Social	Speaker 3 3:15 – 5:00 (105 sec) Live Demo, Phone Sim & Recovery

■■ SPEAKER 1: Problem Statement & 5-Tier Graduated Architecture (0:00 – 1:35)

Goal: Hook the judges, define Problem Statement PS05, explain why static passwords fail, and introduce our sub-10ms 5-tier architecture.

[0:00 – 0:30] The Core Problem:

"Good morning respected judges. In modern cybersecurity, static username and password authentication is completely broken. Attackers deploy automated botnets, credential-stuffing lists from dark-web leaks, and residential proxy networks to impersonate users. However, forcing multi-factor authentication on every single login frustrates legitimate users, driving customer churn."

"DIGNITAS solves Problem Statement PS05: a sub-10 millisecond Adaptive Authentication Gateway that evaluates incoming client telemetry in real time and applies the exact minimum friction needed to protect the account."

[0:30 – 1:10] The 5-Tier Graduated Threat Architecture:

"Rather than a simplistic binary allow-or-block decision, DIGNITAS implements an explainable 5-Tier Risk Engine based on a 0 to 100 risk score:"

Tier	Score	Mitigation Action	Security Objective
Tier 1: Allow	0 – 29	Frictionless Passage (<10ms)	Habitual pattern, trusted device, zero friction
Tier 2: CAPTCHA	30 – 44	Cloudflare Turnstile Proof-of-Work	Neutralizes automated bot velocity & brute force
Tier 3: Step-Up	45 – 59	Phone Prompt (Yes/No) + Email OTP	Enrolls & trusts first-time unrecognized devices
Tier 4: Quarantine	60 – 79	Session Quarantine & Out-of-Band Auth	Quarantines supersonic impossible travel attempts
Tier 5: Hard Block	80 – 100	Block + IP Blacklist + Leaked Alert	Terminates honeytoken hits & credential dumps

[1:10 – 1:35] Speed & Independence:

"Our backend runs on Python FastAPI and SQLite with zero external cloud dependencies. It calculates IP subnet clustering, device fingerprints, and geographic velocity in just 4 to 15 milliseconds. I will now hand over to Speaker 2 to explain how we categorize real-time threats across banking and social applications."

■■ SPEAKER 2: The 5 Threat Vectors & Banking vs Social Policy (1:35 – 3:15)

Goal: Detail the 5 login threat vectors, show our explainable scoring breakdown, and explain why Banking vs Social Media policies differ in risk tolerance.

[1:35 – 2:20] The 5 Threat Vectors in Real Time:

"Thank you Speaker 1. DIGNITAS groups incoming threat signals into five distinct attack vectors:"

1. Credential Replay & Leaks: Detects breached credentials matching dark-web dumps, phishing reverse proxies, and trips honeytoken decoys like our fake backup_admin account.
2. Automated & High-Velocity Threats: Identifies rapid brute-force bursts, credential stuffing from residential proxies, and headless browser automation.
3. Device & Environment Posture: Flags unverified hardware, rooted or jailbroken operating systems, and emulator environments.
4. Network & Location Anomalies: Flags Tor exit nodes, data center proxy subnets, and impossible supersonic travel speeds.
5. Behavioral Anomalies & Session Hijacking: Identifies abnormal login hours, sudden simultaneous attribute shifts, and cross-application correlated attacks.

[2:20 – 3:15] Policy Duality: Banking Zero-Trust vs Social Friction-Balancing:

"A critical innovation of DIGNITAS is Adaptive Policy Profiling. A bank and a social network cannot treat risk the same way:"

Dimension	Apex Bank Profile (Zero-Trust)	Pulse Social Profile (Engagement)

Risk Tolerance	**Low Tolerance**: Prioritizes fraud prevention over convenience	**High Tolerance**: Prioritizes instant onboarding & engagement
Primary Objective	Prevent wire fraud, balance drainage & regulatory fines	Prevent spam botnets, brand hijacking & mass takeovers
Unknown Device	Step-up out-of-band phone prompt; freezes wires for 24h	One-time code; adds to active sessions with alert
Velocity Anomaly	Immediate session quarantine or customer support block	Invisible Turnstile CAPTCHA proof-of-work

"Now Speaker 3 will demonstrate these defenses live using our interactive Gateway and iPhone simulator."

■■ SPEAKER 3: Live Attack Demos, Phone Simulator & Self-Service Recovery (3:15 – 5:00)

Goal: Execute live scenario buttons on screen, show the interactive iPhone simulator buzzing with 'Yes/No' prompts, trigger impossible travel, and demonstrate self-service password recovery.

[3:15 – 3:45] Demo 1: Frictionless Passage vs Bot Defense:

[ACTION: Click Scenario 1 'Familiar Login']

"Watch our live gateway in action. When Alice signs in from her familiar laptop in Delhi, our engine scores it 0 out of 100. Latency is just 6 milliseconds. She receives frictionless access without a single prompt."

[ACTION: Click Scenario 4 'Brute Force Defense']

"Next, an attacker attempts repeated rapid logins with wrong passwords. The engine immediately flags repeated failure velocity, elevates risk to 30 points, and presents a Cloudflare Turnstile CAPTCHA challenge that bot scripts cannot bypass."

[3:45 – 4:25] Demo 2: Phone Simulator Interactive Prompt (Yes/No) & Impossible Travel:

[ACTION: Click Scenario 2 'New Device' & point to Phone Simulator]

"Now, Alice signs in from an unrecognized mobile device. The engine flags Tier 3 (45 pts). Immediately, our iPhone 15 Pro simulator buzzes with an urgent security prompt: Was this you signing in? Notice the two interactive buttons: clicking [✓ Yes, it's me] permanently trusts the device into SQLite and logs her in. Clicking [✕ No, it's not me] instantly emergency-locks the account."

[ACTION: Click Scenario 8 'Impossible Velocity Travel']

"Next, look at Scenario 8: an attacker attempts to log in from London just 15 minutes after Alice was active in Delhi. That requires supersonic aircraft speed. The engine intercepts this as Tier 4 Session Quarantine (70 pts). The user is alerted on their phone with one-click options to [■ Change Password] and [■ Blacklist Attacker IP] directly in SQLite."

[4:25 – 4:45] Demo 3: Honeytoken Trap & Self-Service Forgot Password:

[ACTION: Click Scenario 6 'Honeytoken Trap' then click 'Forgot password?']

"When an intruder probes our decoy backup_admin honeytoken, the engine scores a lethal 100, hard-blocks the connection, and permanently blacklists the IP."

"Finally, we built end-to-end self-service recovery: clicking [Forgot password?] dispatches a 6-digit recovery PIN directly to the phone simulator, allowing the legitimate user to restore access in seconds without needing IT support."

[4:45 – 5:00] Conclusion & Judge Wrap-Up:

"To conclude: DIGNITAS is not a prototype mock-up. It is a production-grade, explainable, sub-10ms adaptive security suite with 100% test accuracy, live smartphone simulation, dual-app threat profiling, and complete deployment readiness. Thank you for your time, and we look forward to your questions!"